Tutorial-2

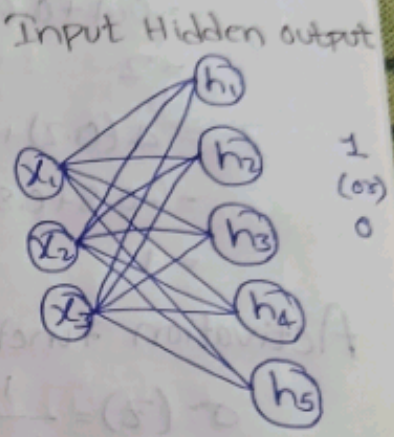
A mobile service provider wants to classify SMS messages as spam or not spam using a single layer neural network with 5 hidden neurons in the layer

x_1 = Specific words and phrases

x_2 = Links and Attachments

x_3 = Bias

x_1	x_2	x_3	Target(y)
4	2	1	1
1	0	1	0
6	6	1	1
2	1	1	0



	W_1	W_2	$b(x_3)$
h_1	0.1	-0.2	1
h_2	0.3	-0.1	1
h_3	0.2	0.4	1
h_4	0.1	0.3	1
h_5	-0.2	0.4	1

$V_1 = 0.1$

$V_2 = 0.2$

$V_3 = -0.3$

$V_4 = -0.1$

$V_5 = 0.4$

Weighted Sum $= x_1 w_1 + x_2 w_2 + b$

$x_1 = 4$

$x_2 = 2$

$b = 1$

$Z_1 = x_1 w_1 + x_2 w_2 + b$

$= 4(0.1) + 2(-0.2) + 1$

$= 1$

$$z_2 = 4(0.3) + 2(-0.1) + 1$$
$$= 1.2 - 0.2 + 1$$
$$= 2$$

$$z_3 = 4(0.2) + 2(0.4) + 1$$
$$= 0.8 + 0.8 + 1$$
$$= 2.6$$

$$z_4 = 4(0.1) + 2(0.3) + 1$$
$$= 0.4 + 0.6 + 1$$
$$= 2$$

$$z_5 = 4(-0.2) + 2(0.4) + 1$$
$$= -0.8 + 0.8 + 1$$
$$= 1$$

Activation function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$h_1 = \sigma(z_1) = \frac{1}{1+e^{-1}} = 0.731$$

$$h_2 = \sigma(z_2) = \frac{1}{1+e^{-2}} = 0.880$$

$$h_3 = \sigma(z_3) = \frac{1}{1+e^{-2.6}} = 0.930$$

$$h_4 = \sigma(z_4) = \frac{1}{1+e^{-2}} = 0.880$$

$$h_5 = \sigma(z_5) = \frac{1}{1+e^{-1}} = 0.731$$

$$z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5$$
$$= 0.73 \times 0.1 + 0.88 \times 0.2 + 0.93 \times (0.3) + (0.88) \times (-0.1) + 0.73 \times 0.4$$
$$= 0.174$$

$$\hat{y} = \sigma(z_{out}) = \frac{1}{1+e^{-0.174}} = 0.543$$

$\sigma(z_{out}) \geq 0.5$

$0.543 \geq 0.5$

$\boxed{\hat{y}=1}$ $\boxed{y=1}$ $\boxed{y=\hat{y}}$

$x_1 = 1, x_2 = 0, b = 1$

$z_1 = x_1 w_1 + x_2 w_2 + b$

$= 1(0.1) + 0(-0.2) + 1$

$= 1.1$

$z_2 = 1(0.3) + 0(-0.1) + 1$

$= 1.3$

$z_3 = 1(0.2) + 0(0.4) + 1$

$= 1.2$

$z_4 = 1(0.1) + 0(0.3) + 1$

$= 1.1$

$z_5 = 1(-0.2) + 0(0.4) + 1$

$= 0.8$

Activation function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$h_1 = \sigma(z_1) = \frac{1}{1+e^{-1.1}} = 0.750$$

$$h_2 = \sigma(z_2) = \frac{1}{1+e^{-1.3}} = 0.785$$

$$h_3 = \sigma(z_3) = \frac{1}{1+e^{-1.2}} = 0.768$$

$$h_4 = \sigma(z_4) = \frac{1}{1+e^{-1.1}} = 0.750$$

$$h_5 = \sigma(z_5) = \frac{1}{1+e^{-0.8}} = 0.689$$

$$Z_{out} = h_1V_1 + h_2V_3 + h_3V_3 + h_4V_4 + h_5V_5$$

$$= 0.750 \times 0.1 + 0.785 \times 0.2 + 0.768 \times (-0.3) + 0.750 \times (-0.1) + 0.689 \times 0.4$$

$$= 0.2022$$

$$\hat{y} = \sigma(z_{out}) = \frac{1}{1+e^{-0.2022}} = 0.550$$

$$\hat{y}(z_{out}) \geq 0.5$$

$$0.550 \geq 0.5$$

$\boxed{\hat{y} = 1}$ But $y = 0$ $y \neq y_p$

Update the weights:

$y = 0$ $\hat{y} = 0.55$ $\eta = 0.1$

$$\eta(y - \hat{y}) = 0.1(0 - 0.55) = -0.055$$

$$V_i^{new} = V_i^{old} + \eta(y - \hat{y})h_i$$

$$V_1^{new} = 0.1 + (-0.055)(0.750) = 0.05875$$

$$V_2^{new} = 0.2 + (-0.055)(0.785) = 0.1568$$

$$V_3^{new} = -0.3 + (-0.055)(0.768) = -0.34224$$

$$V_4^{new} = -0.1 + (-0.055)(0.750) = -0.14125$$

$$V_5^{new} = 0.4 + (-0.055)(0.689) = 0.362105$$

$$Z_{out} = h_1V_1 + h_2V_2 + h_3V_3 + h_4V_4 + h_5V_5$$

$$= 0.150 \times 0.058 + 0.785 \times 0.156 + 0.768 \times (-0.342) + 0.750 \times (-0.141) + 0.689 \times (0.362)$$

$$= 0.0469$$

$$\hat{y} = \sigma(z_{out}) = \frac{1}{1+e^{-0.0469}} = 0.512$$

$\sigma(z_{out}) \geq 0.5$

$0.512 \geq 0.5$

$\boxed{\hat{y}=1}$ But $y=0$ $y \neq \hat{y}$

Update Weights:

$$v_i^{new} = v_i^{old} + \eta(y-\hat{y})h_i$$

$y=0$ $\hat{y}=0.512$ $\eta=0.1$

$$\eta(y-\hat{y}) = 0.1(0-0.572) = -0.0512$$

$$v_1^{new} = 0.1+(-0.0572)(0.750) = 0.061$$

$$v_2^{new} = 0.2+(-0.0572)(0.785) = 0.159$$

$$v_3^{new} = -0.3+(-0.0572)(0.768) = -0.339$$

$$v_4^{new} = -0.1+(-0.0572)(0.750) = -0.138$$

$$v_5^{new} = 0.4+(-0.0572)(0.689) = 0.364$$

$$z_{out} = h_1v_1+h_2v_2+h_3v_3+h_4v_4+h_5v_5$$

$$= 0.750\times0.061+0.785\times0.159+0.768\times-0.33+0.750\times-0.138+0.689\times0.364$$

$$= 0.0575$$

$$\hat{y} = \sigma(z_{out}) = \frac{1}{1+e^{-0.0575}}$$

$$= 0.5143$$

$z_{out} \geq 0.5$ $(z_{out}) \leq 0.5$

$\boxed{0.5143 \geq 0.5}$

$\hat{y}=1$ $y=0$ $y \neq \hat{y}$

$x_1=6$ $x_3=3$ $b=1$

$$z_1 = x_1w_1+x_2w_2+b$$

$$= 6(0.1)+3(-0.2)+1 = 1$$

$Z_2 = 6(0.3) + 3(-0.1) + 1 = 2.5$

$Z_3 = 6(0.2) + 3(0.4) + 1 = 3.4$

$Z_4 = 6(0.1) + 3(0.3) + 1 = 2.5$

$Z_5 = 6(-0.2) + 3(0.4) + 1 = 1$

Activation Function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$h_1 = \sigma(z_1) = \frac{1}{1+e^{-1}} = 0.731$$

$$h_2 = \sigma(z_2) = \frac{1}{1+e^{-2.5}} = 0.924$$

$$h_3 = \sigma(z_3) = \frac{1}{1+e^{-3.4}} = 0.967$$

$$h_4 = \sigma(z_4) = \frac{1}{1+e^{-2.5}} = 0.924$$

$$h_5 = \sigma(z_5) = \frac{1}{1+e^{-1}} = 0.731$$

$$Z_{out} = h_1 V_1 + h_2 V_2 + h_3 V_3 + h_4 V_4 + h_5 V_4$$

$$= 0.731(0.1) + 0.924(0.2) + 0.967(-0.3) + 0.924(-0.1) + 0.731(0.4)$$

$$= 0.1678$$

$$\hat{y} = \sigma(Z_{out}) = \frac{1}{1+e^{-0.1678}}$$

$$= 0.541$$

$$\sigma(Z_{out}) \geq 0.5$$

$$0.541 \geq 0.5$$

$\boxed{\hat{y} = 1}$ $\boxed{y = 1}$ $\boxed{y = \hat{y}}$

$x_1 = 2$ $x_2 = 1$ $b = 1$

$$Z_1 = x_1 w_1 + x_2 w_2 + b$$

$$= 2(0.1) + 1(-0.2) + 1 = 1$$

$$Z_2 = 2(0.3) + 1(-0.1) + 1 = 1.5$$

$z_3 = 2(0.2)+1(0.4)+1 = 1.8$

$z_4 = 2(0.1)+1(0.3)+1 = 1.5$

$z_5 = 2(-0.2)+1(0.4)+1 = 1$

Activation Function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$h_1 = \sigma(z_1) = \frac{1}{1+e^{-1}} = 0.731$$

$$h_2 = \sigma(z_2) = \frac{1}{1+e^{-1.5}} = 0.817$$

$$h_3 = \sigma(z_3) = \frac{1}{1+e^{-1.8}} = 1.858$$

$$h_4 = \sigma(z_4) = \frac{1}{1+e^{-1.5}} = 0.817$$

$$h_5 = \sigma(z_5) = \frac{1}{1+e^{-1}} = 0.731$$

$$z_{out} = h_1v_1 + h_2v_2 + h_3v_3 + h_4v_4 + h_5v_5$$

$$= 0.731(0.1) + 0.817(0.2) + 0.858(-0.3) + 0.817(-0.1) + 0.731(0.4)$$

$$= 0.1898$$

$$\hat{y} = \sigma(z_{out}) = \frac{1}{1+e^{-1.898}} = 0.547$$

$$\sigma(z_{out}) \geq 0.5$$

$$0.547 \geq 0.5$$

$\hat{y} = 1$ $\quad y = 0$ $\quad y \neq \hat{y}$

Update Weights:

$$v_i^{new} = v_i^{old} + \eta(y-\hat{y})h_i$$

$y = 0$ $\quad \hat{y} = 0.547$ $\quad \eta = 0.1$

$$\eta(y-\hat{y}) = 0.1(0-0.547)$$

$$= -0.547$$

$$v_1^{new} = 0.1 + (-0.0547)(0.731) = 0.060$$

$$v_2^{new} = 0.2 + (-0.0547)(0.817) = 0.155$$

$V_3^{new} = -0.3 + (-0.0547)(0.858) = -0.346$

$V_4^{new} = -0.1 + (-0.0547)(0.817) = -0.144$

$V_5^{new} = 0.4 + (-0.0547)(0.731) = 0.360$

$Z_{out} = h_1V_1 + h_2V_2 + h_3V_3 + h_4V_4 + h_5V_5$

$= 0.731 \times 0.060 + 0.817 \times 0.155 + 0.858 \times (-0.346) + 0.817 \times (-0.144) + 0.731 \times 0.366$

$= -0.222$

$\hat{y} = \sigma(Z_{out}) = \frac{1}{1+e^{(0.222)}}$

$= 0.444$

$\sigma(Z_{out}) \geq 0.5$

$0.444 \geq 0.5 \qquad 0.444 \leq 0.5$

$\boxed{\hat{y}=0} \qquad \boxed{y=0} \qquad \boxed{y=\hat{y}}$